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Atlantic Technological University

GaitSyncAR

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GitHub:

Introduction

Gait impairments from conditions like Stroke and Meningitis significantly hinder independence. Traditional rehabilitation is often labour intensive and lacks quantitative feedback.

The Motivation: Inspired by our project owner who previously used a metronome strapped to a bicycle helmet to help synchronise their gait rhythm.

GaitSyncAR modernises rehabilitation by transposing rhythmic cueing into an immersive Augmented Reality (AR) environment, paired with haptic and audio feedback, as well as gait analysis from the ankle sensors.

The Challenge: Most AR interfaces utilise gesture-based controls inaccessible to those with tremors. This project prioritises **Accessible UI/UX** and safety through **Optical See-Through (OST)** technology.

Methodology

Phase 1: Research & Requirement Analysis

- AR Technology Evaluation.
- Gait Rehabilitation Techniques.

Phase 2: Sequential Hardware Engineering (Waterfall)

- **Topology Design:** Developed schematics for sensors housing and circuits.
- **Circuit Simulation:** Validated circuit logic via Falstad circuit simulator.
- **Breadboard Prototyping:** Assembled a physical prototype to test component functionality with a Digital Multimeter.
- **Final Implementation:** Transitioned from solderless connections to a robust Perfboard build for final system integration.

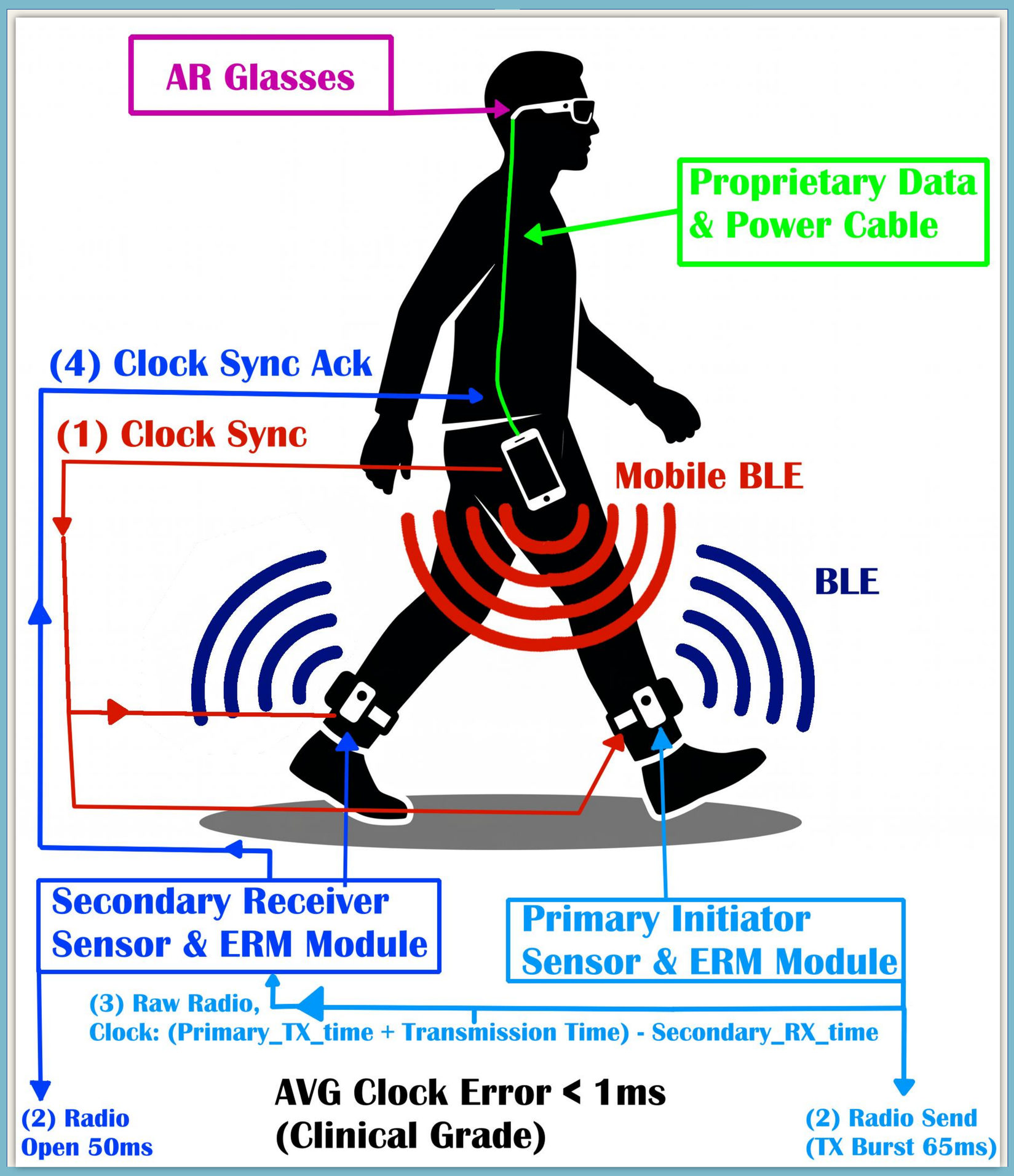
Phase 3: Iterative Software Development

- Software was continuously iterated on and tested in a loop lifecycle.
- **User Testing**: Conducted regular feedback sessions with the project owner to refine features & accessibility.

Conclusion

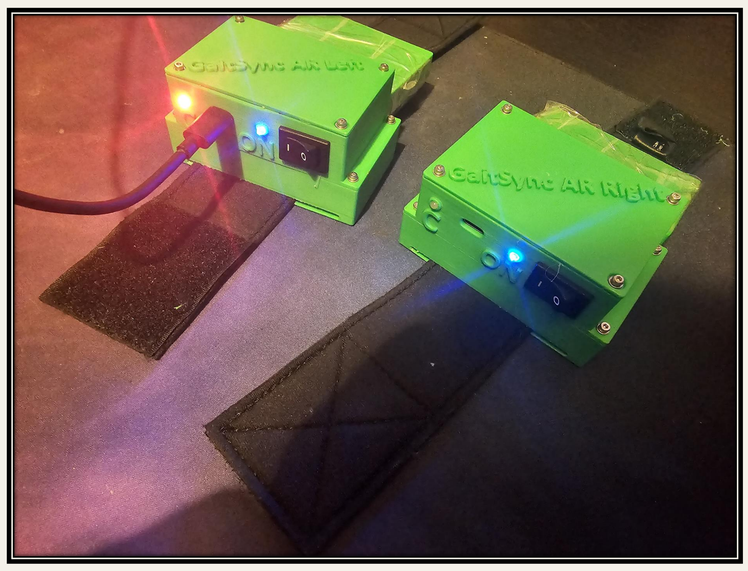
Key Outcomes

- **Seamless Integration**: Validated zero-latency "handshake" between mobile device and ankle sensors.
- **Accessibility First:** First-hand feedback that gesture-free interface is highly effective for users with motor impairments or tremors.
- **Clinical Validation:** The primary participant showed immediate, significant improvement in walking rhythm while using the device. Crucially, a "carry-over" effect was observed, with improved gait fluidity persisting shortly after the device was removed.

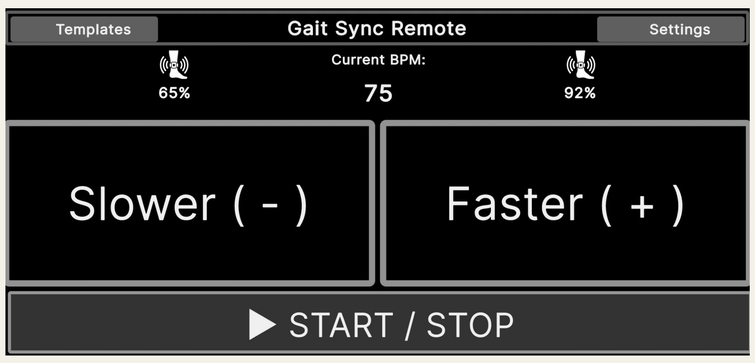


System Design

Ankle Sensors:



Unity Interface



Technologies

Software:

- **Augmented Reality (AR)**
- **Unity (C#):** Rendering 3D AR environment.
- **System Bridge:** Unity-to-AR C Bridge manages system permissions and translates real-time sensor data into the AR environment.

Hardware:

- The system is powered by the **Seeed Studio XIAO nRF52840** microcontroller.
- **Motion Tracking:** utilises the on-board 6-axis IMU to capture high-fidelity gait movement data.
- **Real-Time OS**: Built on Zephyr **(C)**, ensuring deterministic execution for precise rhythmic synchronisation and data collection.
- **Low-Latency Communication, Dual-Stack Connectivity:** Bluetooth Low Energy (BLE) manages data exchange between the wearable and the AR interface.
- **Bare Metal:** Custom raw radio protocols are implemented to achieve sub-1-millisecond clock synchronisation between the two ankle sensors.
- **Additive Manufacturing: 3D Printing** was utilised to develop custom enclosures for the wearable hardware.